NIELS REIJERS

Staff / Lead Software Engineer

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Full CV

Staff / Lead Software Engineer specialising in distributed systems, backend architecture, performance engineering and AI agent infrastructure.
Experienced in taking complex systems from design to production, improving reliability and scaling critical services.

PhD in Computer Science, with research published at SenSys 2018.

EXPERIENCE

Senior Software Engineer
Aug 2024 – Present
Proton Mail
Taipei, Taiwan

Lumo — AI platform (2026–Present)
- Develop and scale Rust backend services for Proton's AI offering.
- Lead the effort to deliver direct API access to Lumo.
- Initiated user-facing features: background generation, an "Answer Now" control to prioritise faster responses, and improved user-feedback collection.

AI Agents (2025)
- Shaped the architecture and prompt design of an AutoGen-based framework for LLM-driven agentic task automation.

DDoS protection and traffic analysis (2024–2025)
- Increased DDoS protection capacity 5× by isolating real-time traffic analysis from resource-intensive analytics queries.
- Improved root-cause analysis for traffic anomalies, reducing time to diagnosis.

Rust | LLM infrastructure | vLLM / sglang | ClickHouse | Python

Staff Software Engineer
Jun 2011 – Apr 2024
DSW Health Insurance (contract)
Schiedam, The Netherlands

Continued as a contract engineer while pursuing doctoral research in Taiwan; from 2016 as Staff Software Engineer.
- Designed and delivered multiple production systems; served as a go-to troubleshooter across teams.
- Drove improvements in automation, observability, and performance.
- Initiated high-impact engineering projects, including:
  - Company-wide service observability, providing real-time insight into system behaviour.
  - A move from script-based deployments to a convention-over-configuration system, improving deployment reliability and developer efficiency.

C# | MS SQL Server

Postdoctoral Researcher
Feb 2019 – May 2021
National Taiwan University / Academia Sinica
Taipei, Taiwan

Doctoral Candidate
Sep 2011 – Apr 2018
Intel-NTU Connected Context Computing Center
Taipei, Taiwan

Research focused on improving Java virtual machines for resource-constrained IoT devices.
- Developed on-device ahead-of-time compilation to native code, bringing performance close to optimised C while reducing energy consumption.
- Designed a platform-independent sandboxed environment to protect devices from malicious code.

Results published at ACM SenSys 2018 (top-tier systems conference).

C | Embedded systems

Lead Developer
Apr 2005 – Nov 2009
DSW Health Insurance
Schiedam, The Netherlands

Led development of DSW's new claims-processing system.
- Delivered the system on schedule. It has run reliably since 2009 and remains maintainable despite being the company's most complex system.
- Developed spin-off products that became company-wide standards, improving efficiency and standardisation across teams.

SKILLS

AI: LLM infrastructure, prompt engineering, agent frameworks (AutoGen)

Languages: Rust, Python, C#, Java, Go, C and AVR assembly

Data & Storage: MS SQLServer, ClickHouse, MySQL, Datadog Vector

Systems: Distributed systems, observability, performance optimization

EDUCATION

Ph.D. in Computer Science
National Taiwan University
2011 – 2018
Taipei, Taiwan

"CapeVM: A Fast and Safe Virtual Machine for Resource-Constrained Internet-of-Things Devices", graded A+.

M.Sc. in Computer Science
Delft University of Technology
1995 – 2002
Delft, The Netherlands

"Location tracking and group communication in FLARE", graded 9/10.

Research done at the Distributed Systems Group in Trinity College Dublin.

LANGUAGES

Dutch ●●●●●
English ●●●●●
Mandarin Chinese ●●●○○

INTERESTS

- Cycling
- Volunteer at the Rotterdam International Film Festival, before moving to Taipei
- Led the winning company team in the 2013 Delft University of Technology programming contest